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1 Executive Summary

In the field of cyber security particularly in ethical hacking. It is very Important to work in safe and isolated network environment before initiating a security assessment against any target. This is important to avoid unintended attacks on live production system other than that this is also important when dealing with malware analysis this isolation will help you to prevent your network and data from malicious actors this report present how to setup a professional and secure lab environment for ethical hacking.

2 Scope of Task

The Scope of this task is to build a safe virtual environment for penetration testing. Install and Configure Oracle virtual box. Import and run the kali linux virtual machine after then verify that the system and network is correctly working

3 Tools Used

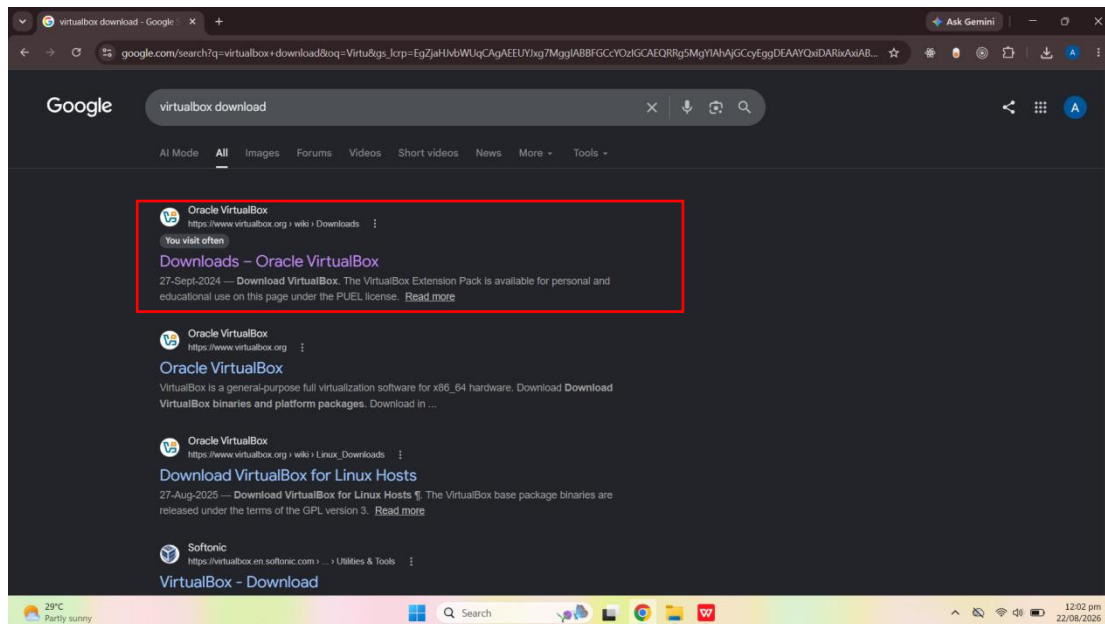
1. Virtual box
2. Host Operating System windows 11
3. Kali Linux (Attacker Machine)

4 Downloading and Installing Virtual box

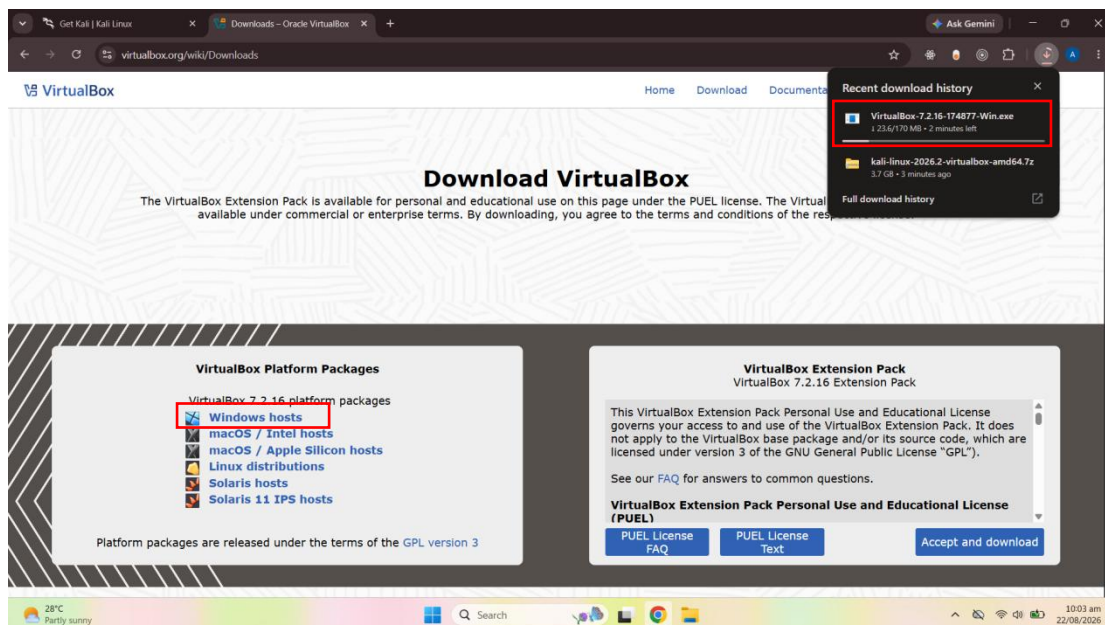
4.1 Downloading Virtual box

To Download the virtual box follow the steps given below

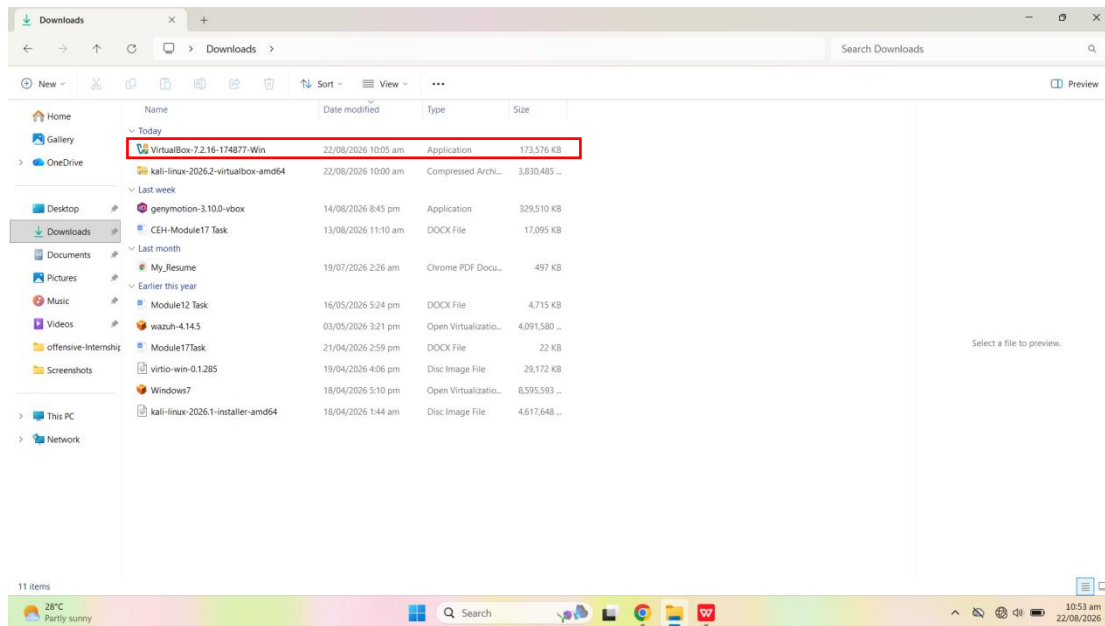
- 1) Go to Google.com and Search Virtual box Download select the official website
Link: <https://www.virtualbox.org>



- 2) Open a website Select the virtual box download according to your host operating system as my Host operating system is windows so I selected windows.



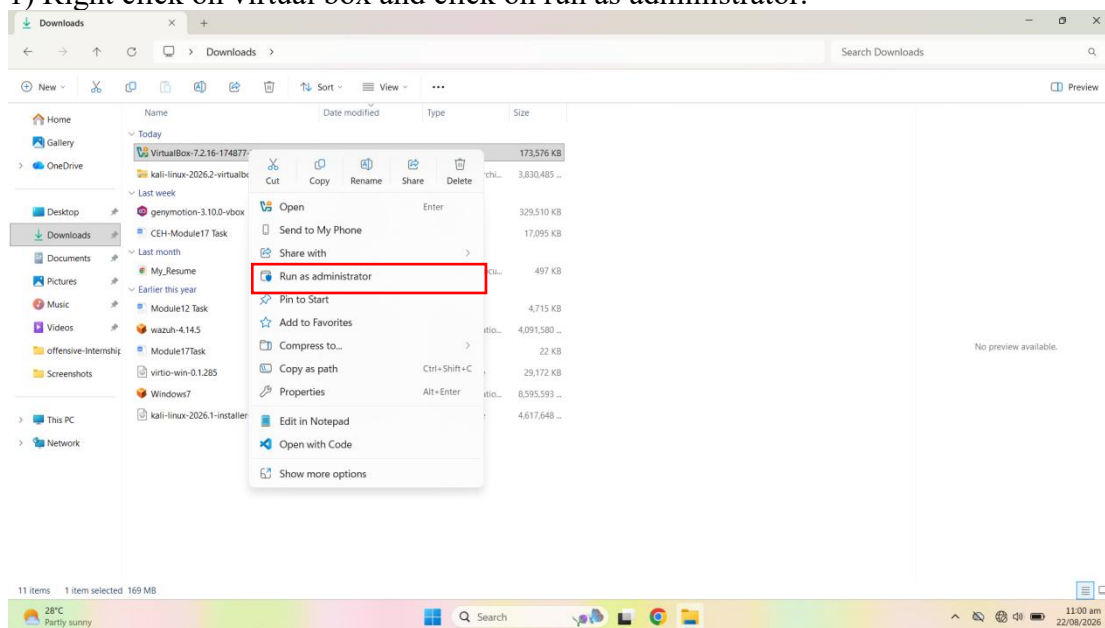
- 3) As Virtual box is successfully Downloaded can be seen in Downloads folder.



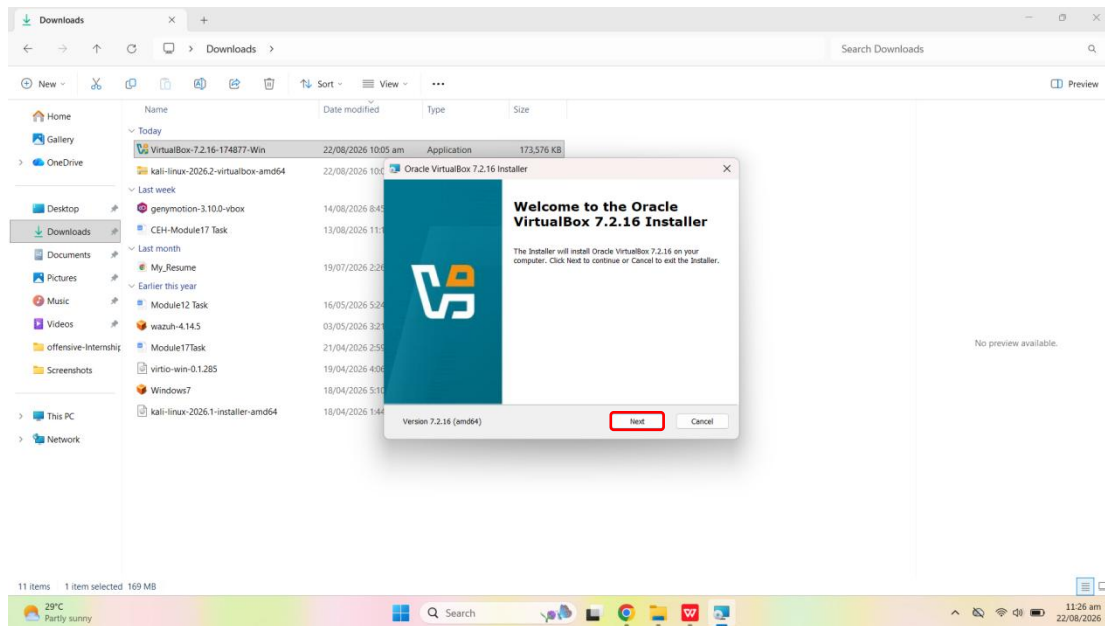
4.2 Installing Virtual box

After downloading follow the steps given below to install virtual box in a System.

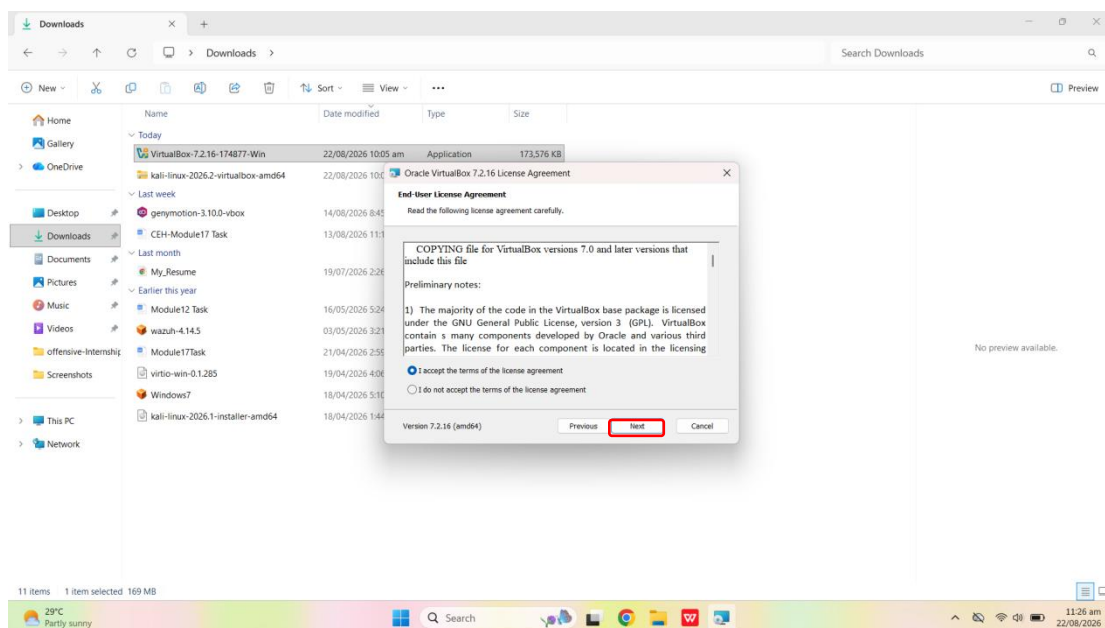
1) Right click on virtual box and click on run as administrator.



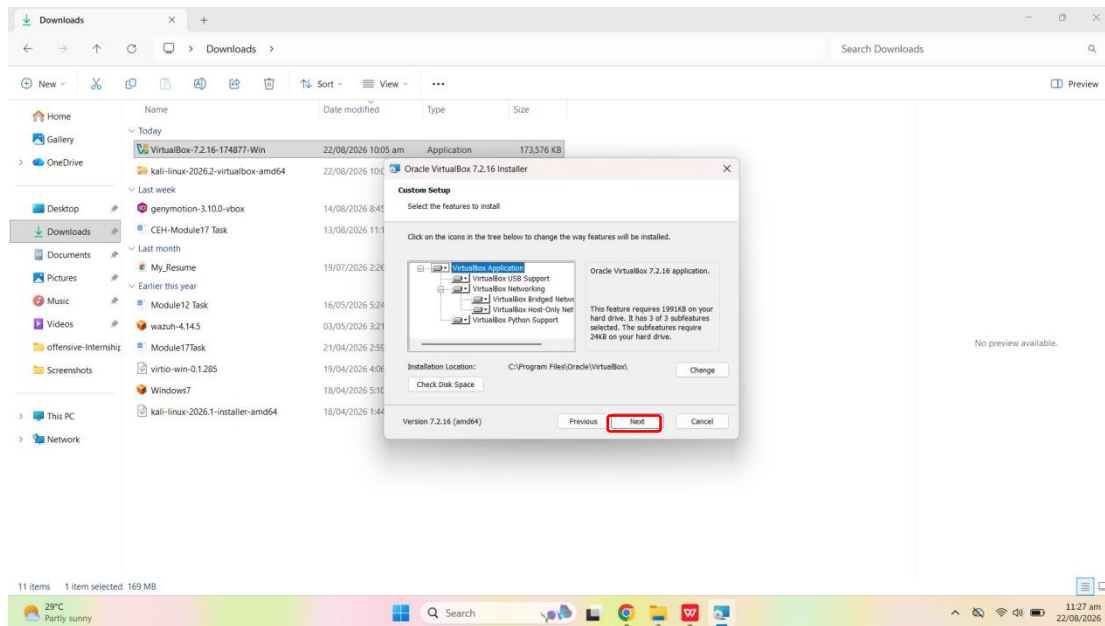
2) Click on Next



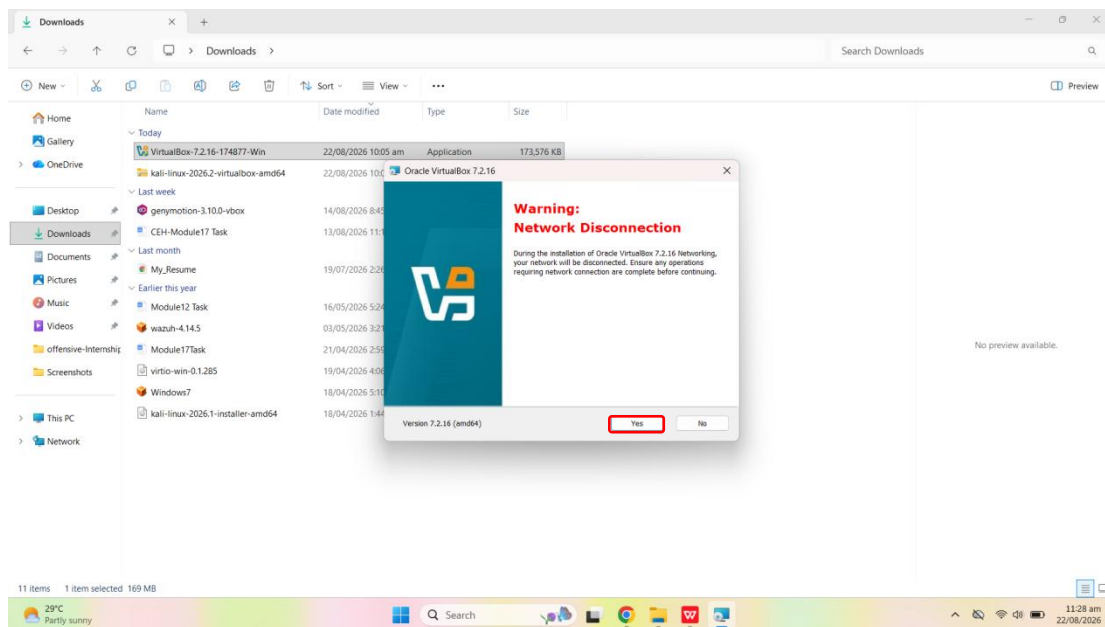
3) Click on I accept the terms of license agreement and then click next.



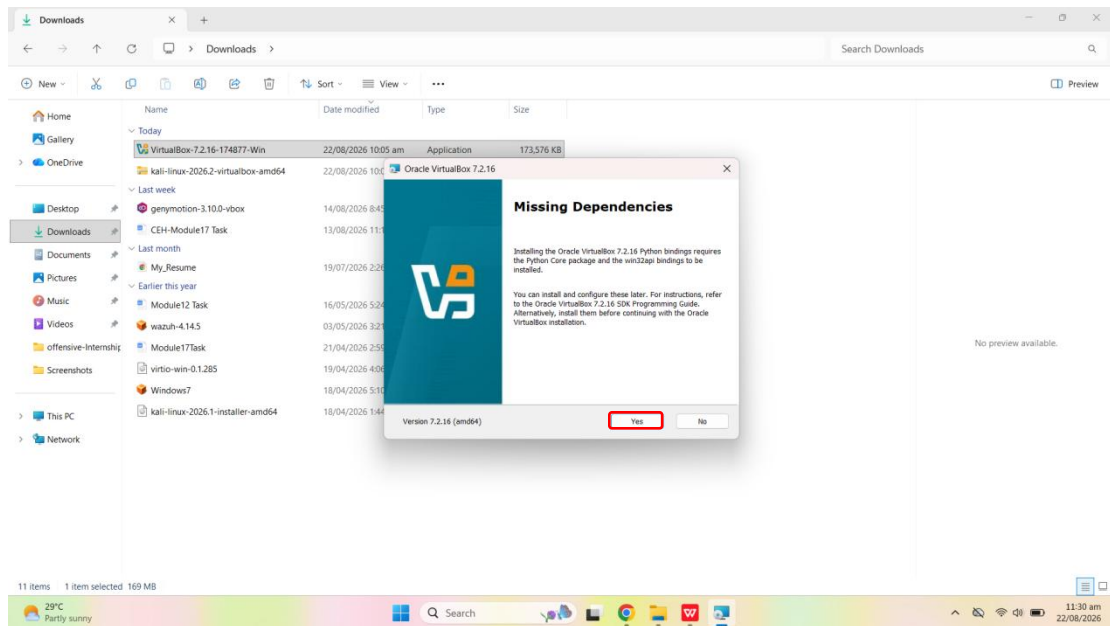
4) As default location of virtual box will be in C drive simply Press next.



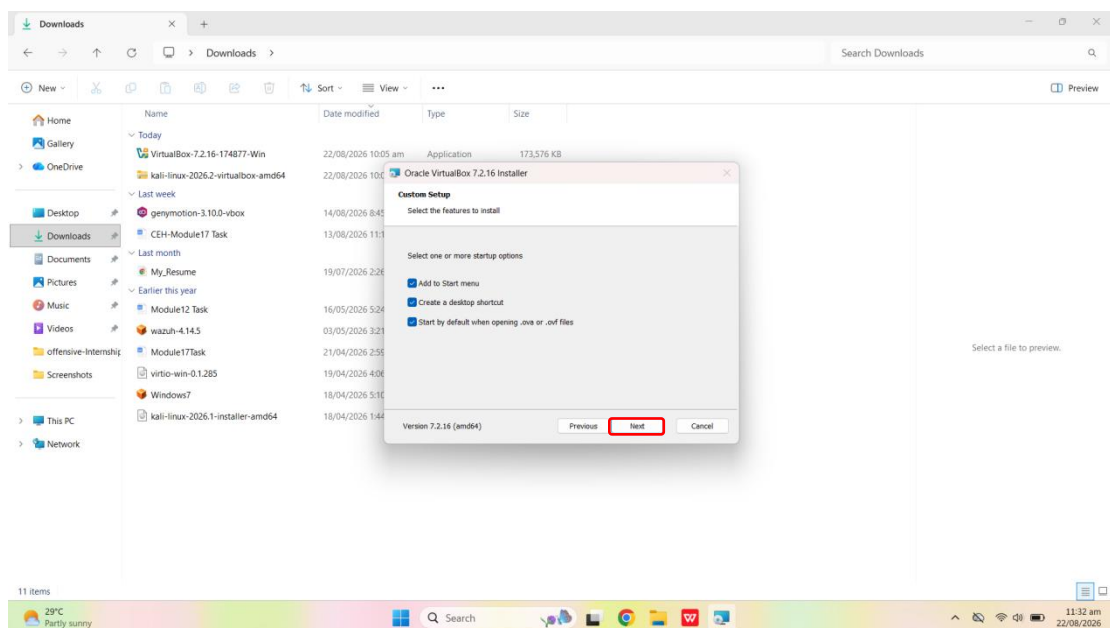
5) During the Installation there may be a slightly network interruption for a while.
Click next



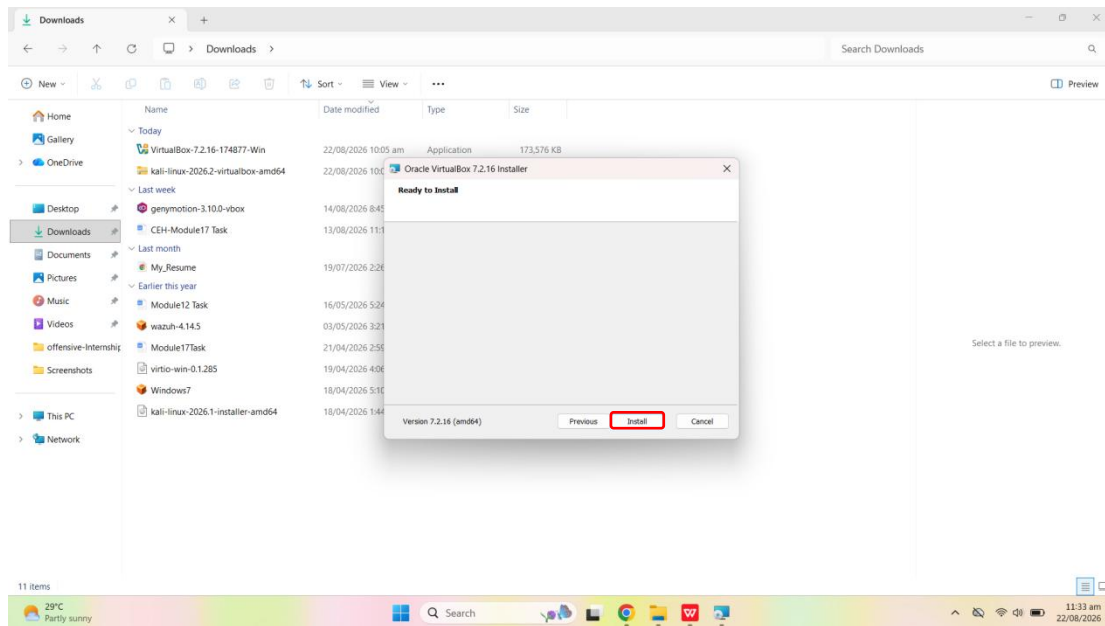
6) Click Yes



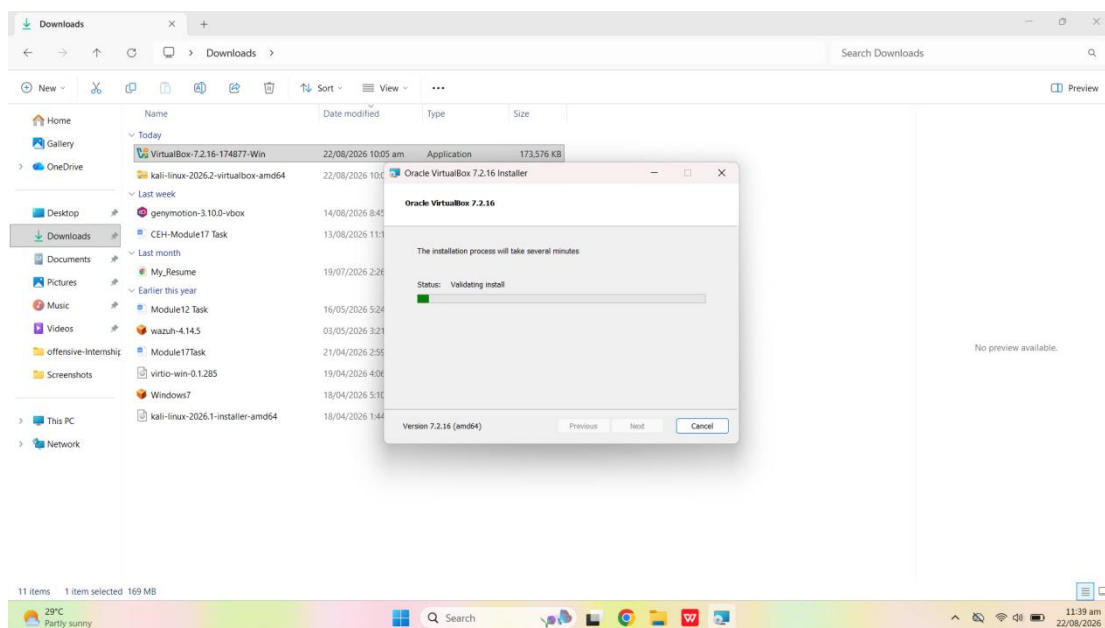
7) Click Next



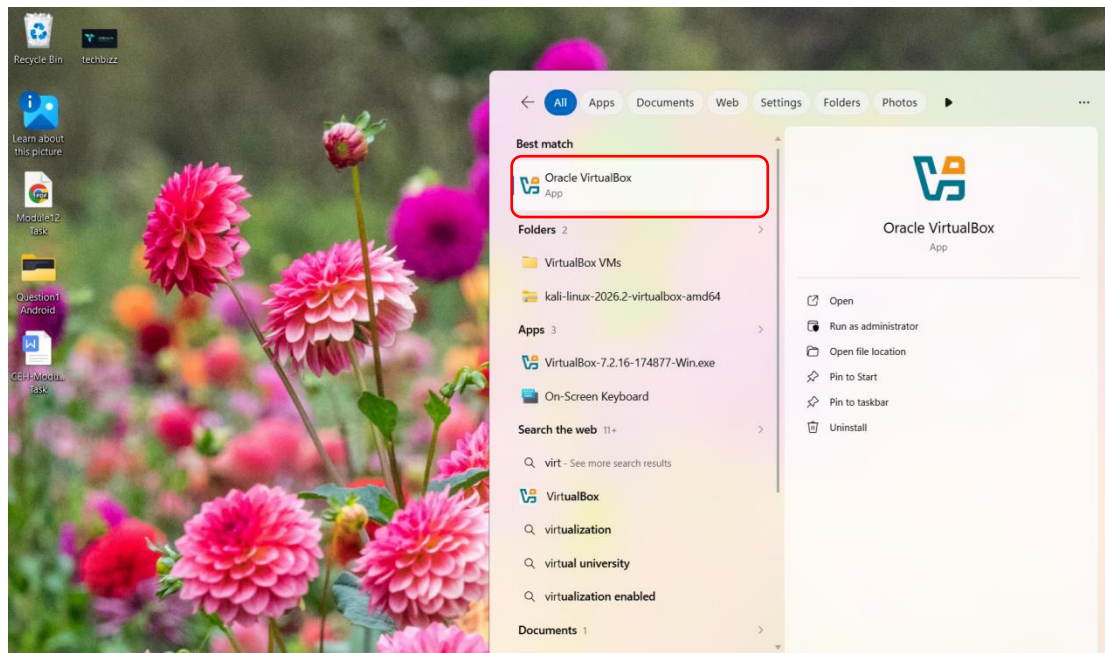
8) Click Install



9) Now installation is in Progress wait for few minutes.



10) Virtual box is successfully downloaded.



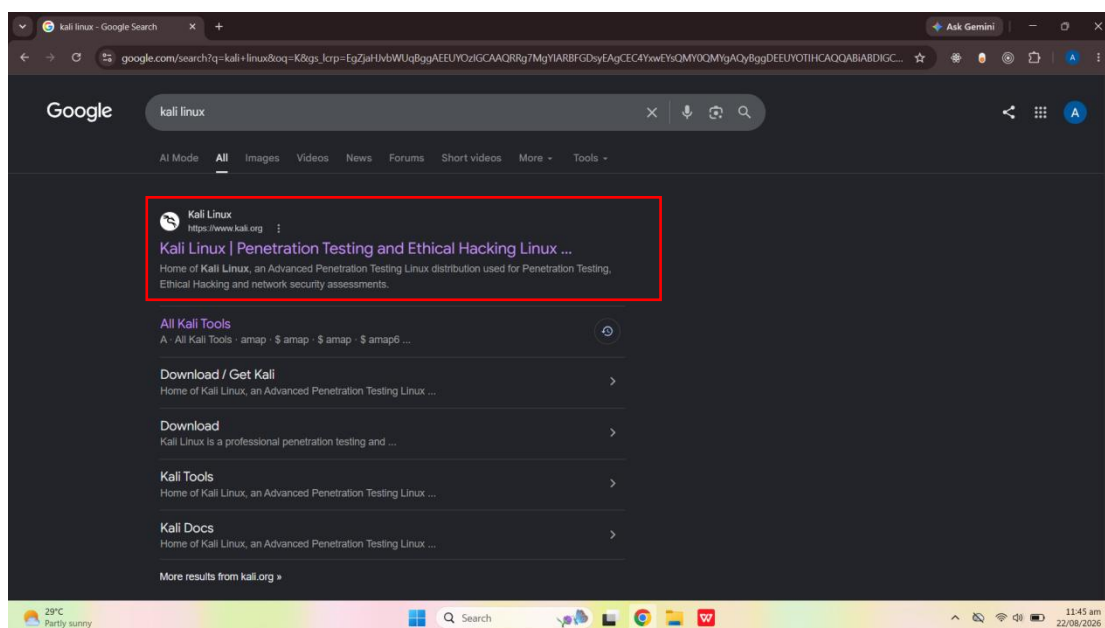
5 Downloading and Installing Kali Linux

5.1 Downloading Kali Linux

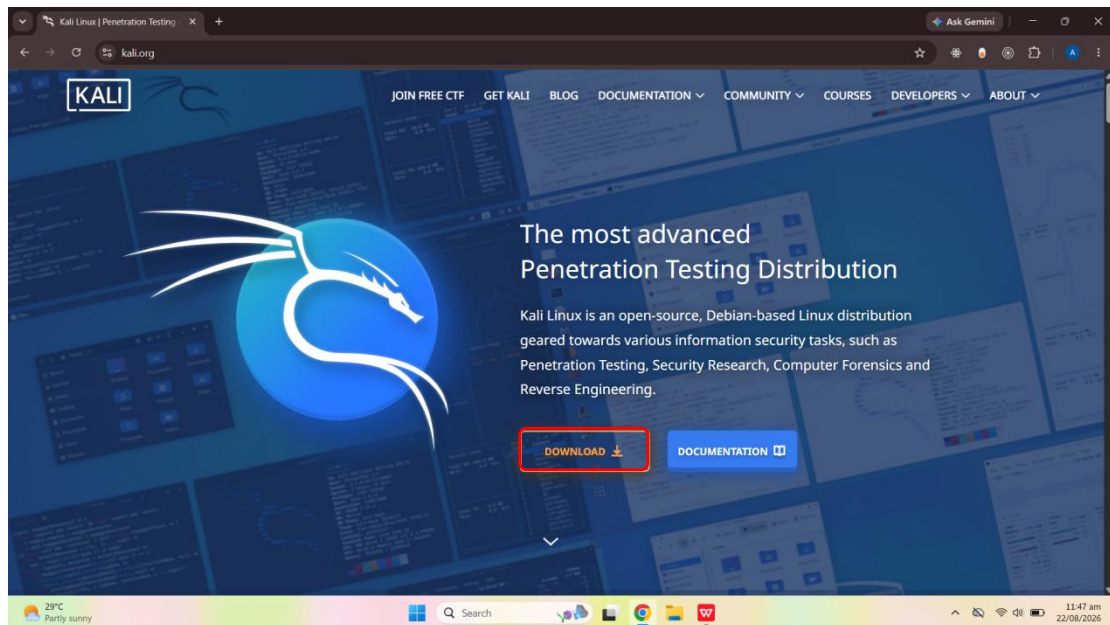
Downloading Kali Linux has 2 options Installer Images and Virtual box Images. Virtual Box Images is a easy way because kali Linux is Pre-built vm means all the configurations are already set by default no need to install and configure anything.

To Download kali linux Follow the steps given below

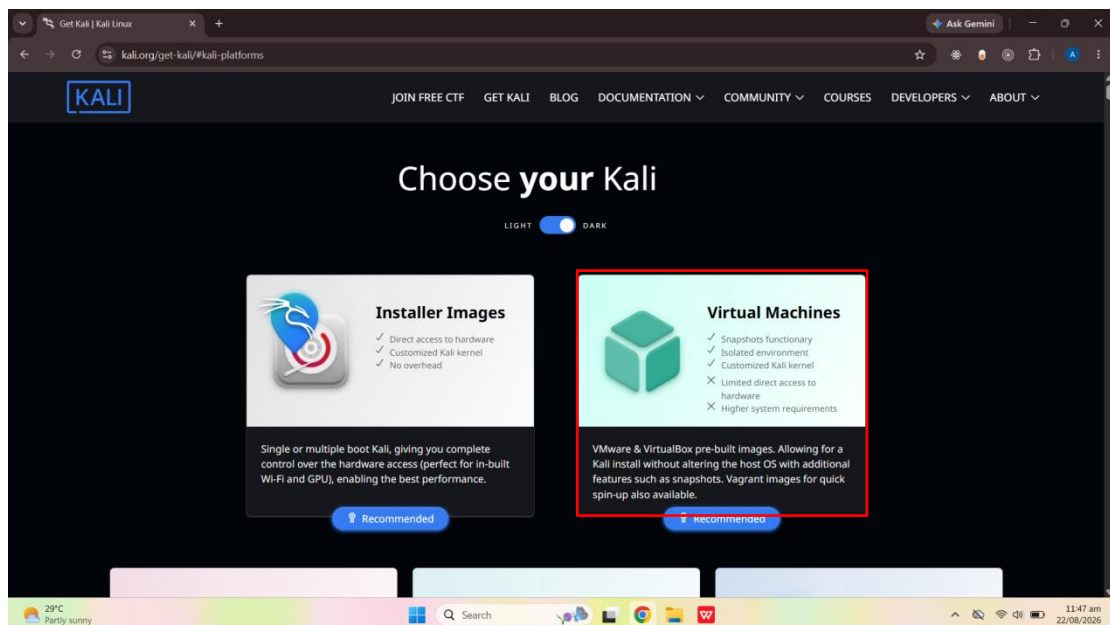
1. Go to google.com and Search Kali Linux select the official Kali linux website shown in Image below. <https://www.kali.org>



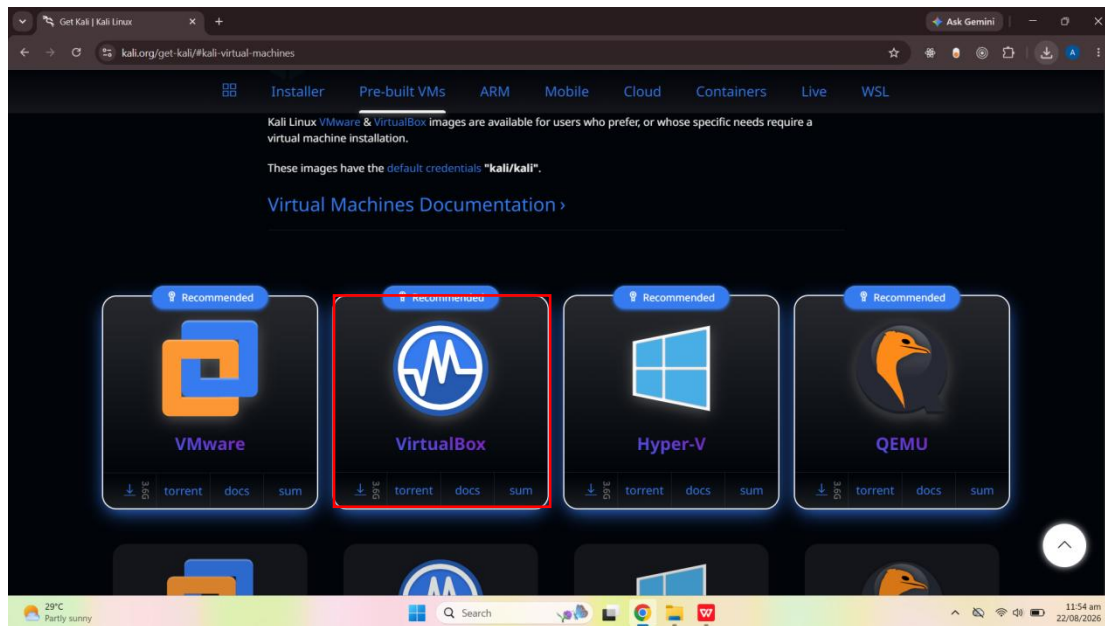
2. After visiting the website click on Download



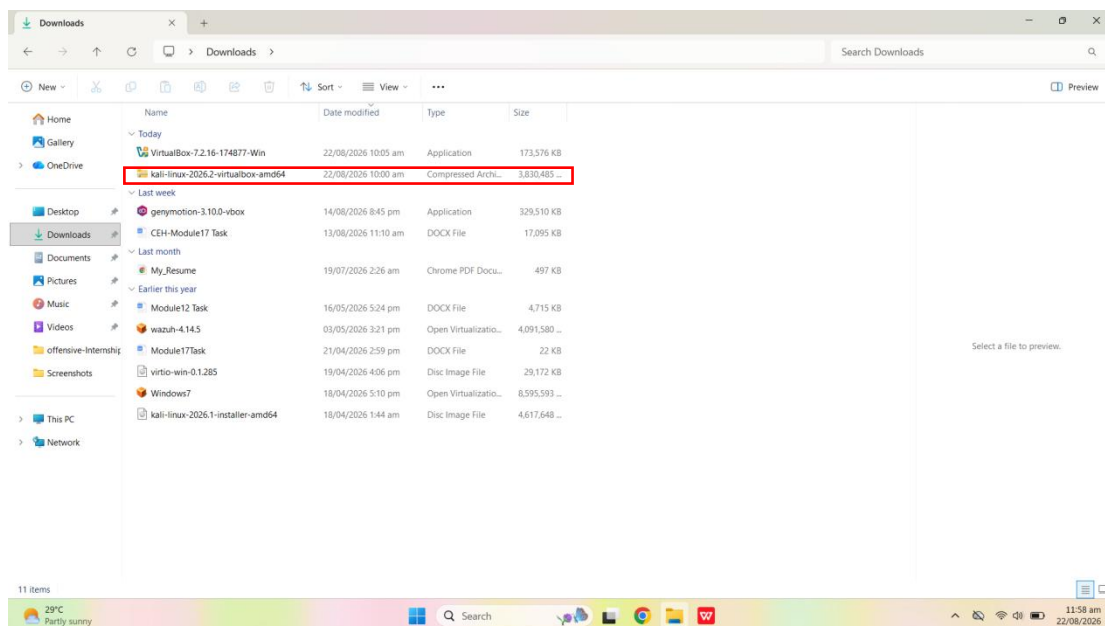
3. Select the Virtual machines.



4. Click on the Virtual box as shown in Image below to download It will take some time depending upon the internet Speed



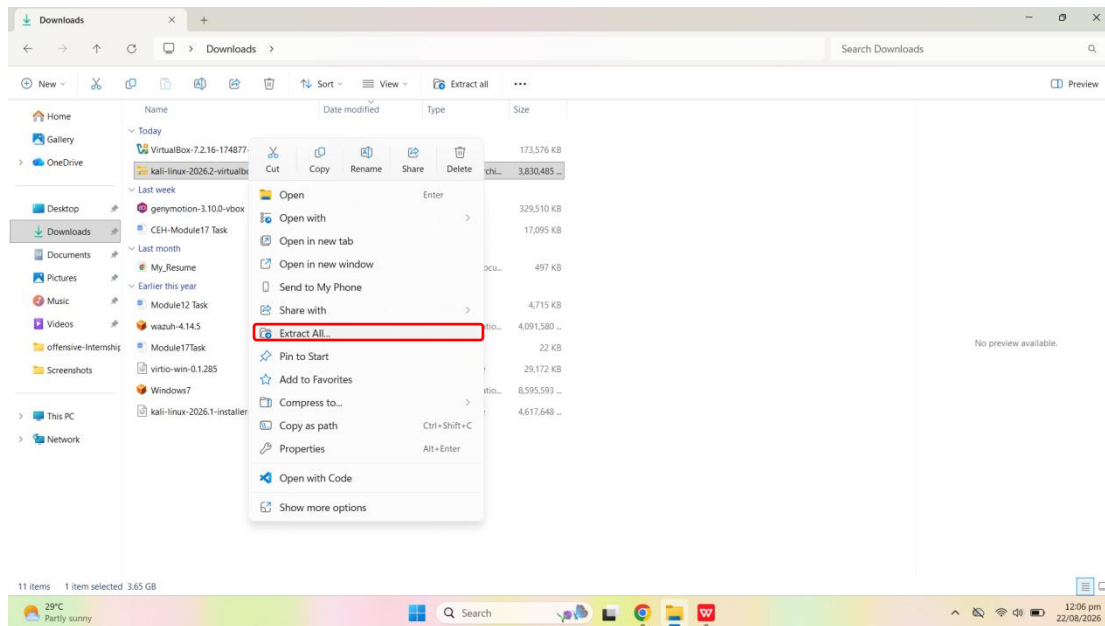
- As kali Linux Pre-built Vm is downloaded in System Successfully can be shown in Downloads folder.



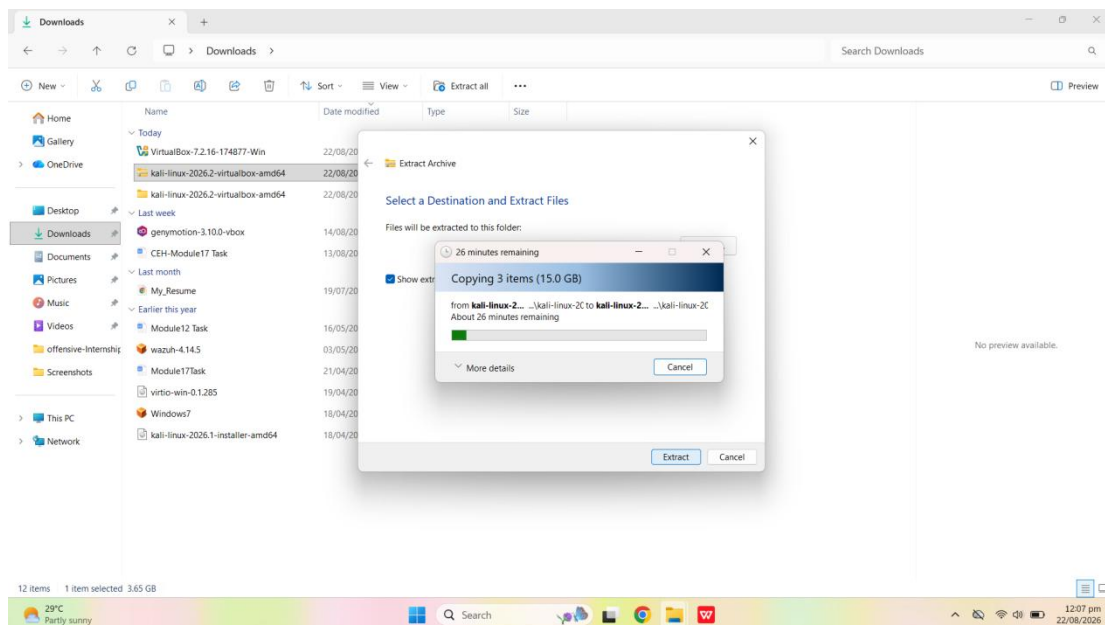
5.2 Installing Kali Linux

To Install Kali linux Follow the steps given below

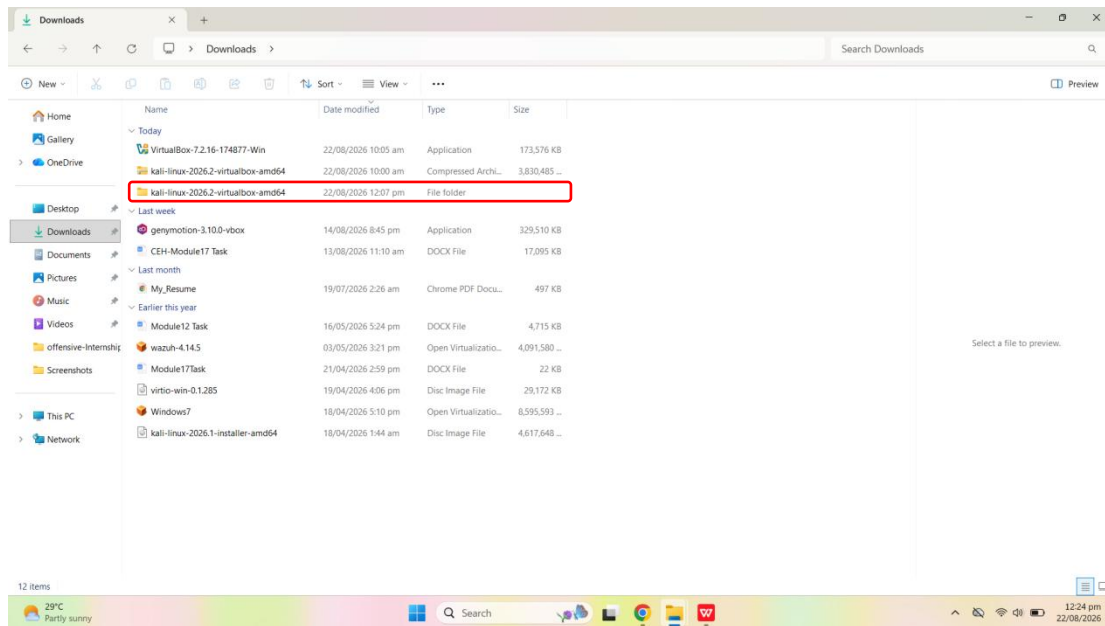
- Right click on the zipped folder of kali linux and Click on Extract all



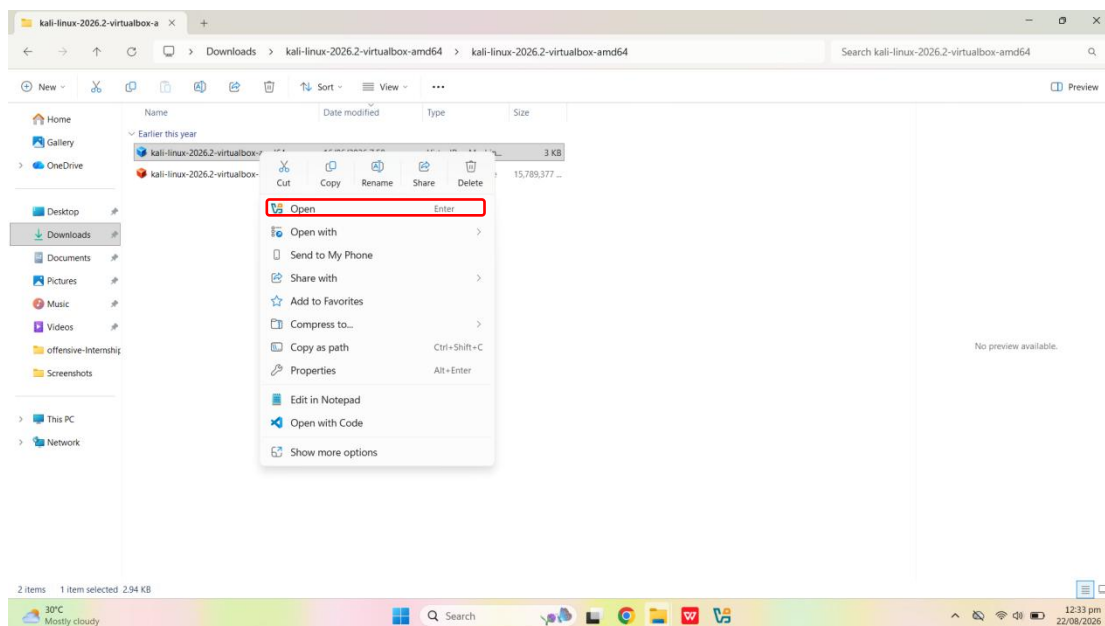
- 2) Select Destination and then Press Extract Extraction Process is in Continue wait for few minutes



- 3) As Extraction is completed Successfully



- 4) Open the Extracted folder and click on file pre built vm of kali linux will be added in virtual box.

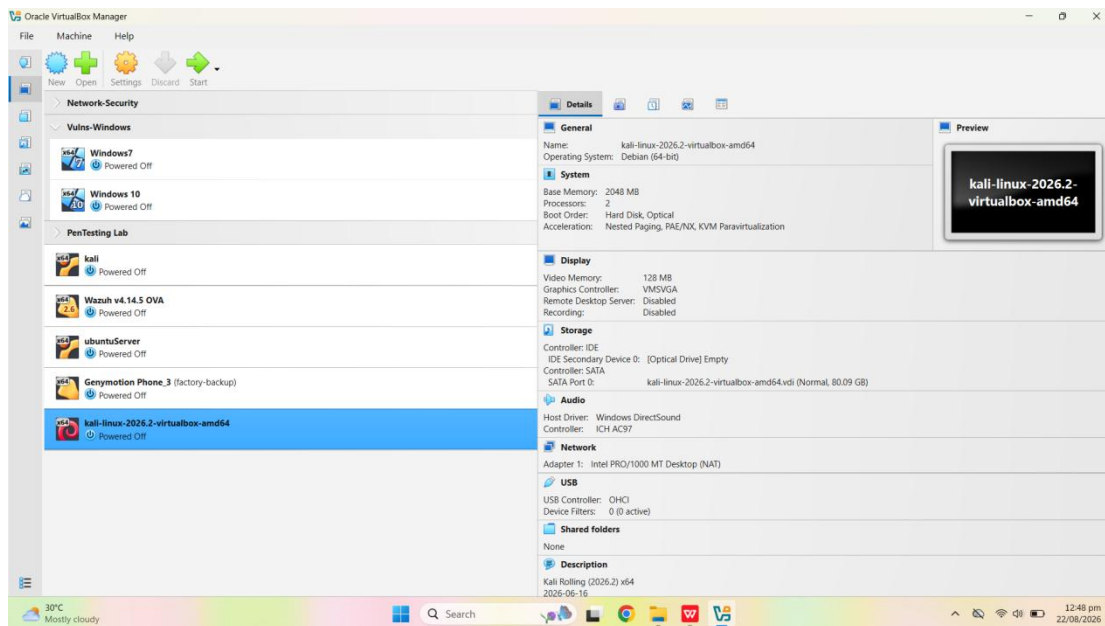


- 5) Kali Linux machine is successfully imported in oracle virtual box. Now its time to check the assigned hardware resources before booting the kali.

Virtual Machine Configuration

1. **Operating System:** Debian-based
2. **Ram:** 2GB
3. **Processors:** 2 Cpu Cores
4. **Storage:** 80GB
5. **Network Adapter:** NAT

These are the default Configuration of Kali Virtual Machine we can adjust hardware resources as per our System's specification for better performance.



As Kali Linux is successfully booted up know



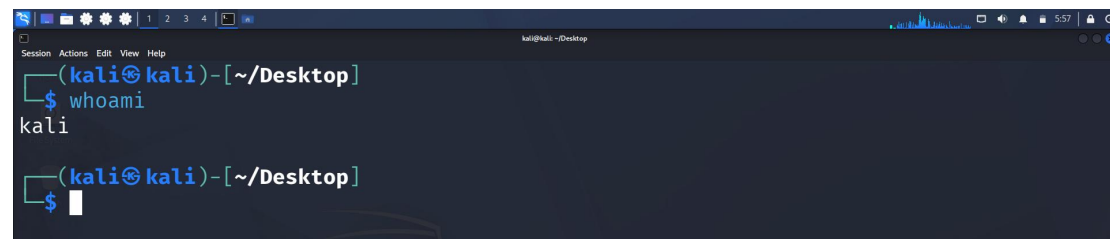
Open the Terminal and Run the commands one by one to check user account and Network configuration is working properly Commands are mentioned below

- whoami
- ip a
- sudo apt update

6 Basic Kali Linux Commands

6.1 Command whoami

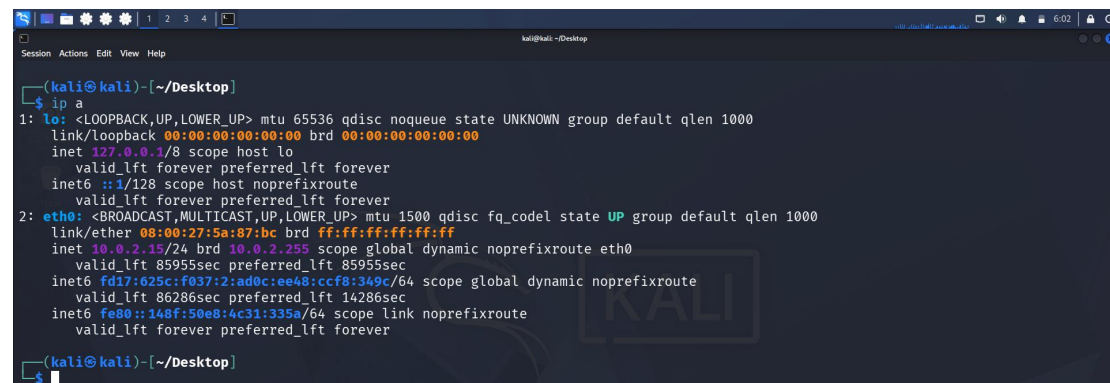
This command is use to check the current logged-in user.



```
(kali㉿kali)-[~/Desktop]
$ whoami
kali
(kali㉿kali)-[~/Desktop]
$
```

6.2 Command ip a

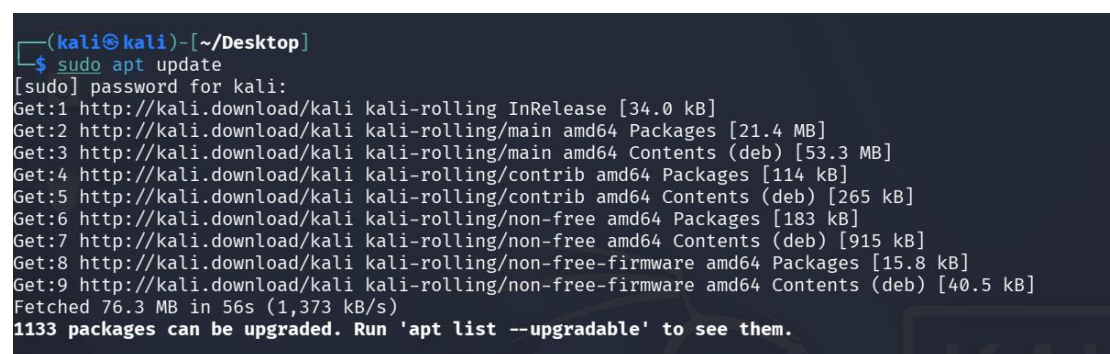
This command is use to check the ip address of a system.



```
(kali㉿kali)-[~/Desktop]
$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
   link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
   inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
   inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
   link/ether 08:00:27:5a:87:bc brd ff:ff:ff:ff:ff:ff
   inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
       valid_lft 85955sec preferred_lft 85955sec
   inet6 fd17:625c:f037:2:ad0c:ee48:ccf8:349c/64 scope global dynamic noprefixroute
       valid_lft 86286sec preferred_lft 14286sec
   inet6 fe80::148f:50eb:4c31:335a/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
(kali㉿kali)-[~/Desktop]
$
```

6.3 Command sudo apt update

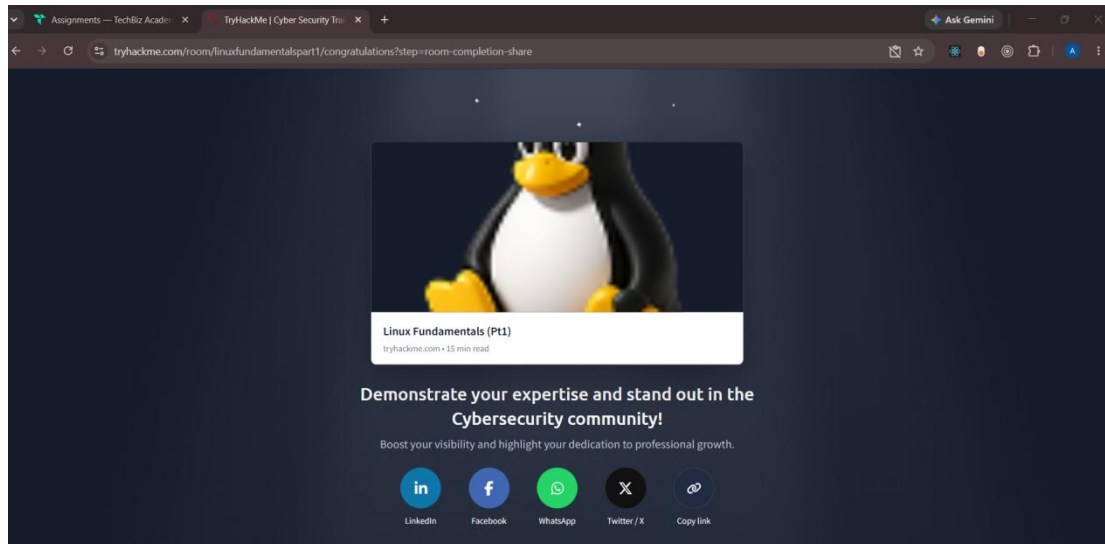
This command is use to update all the packages from the online repositories inside system so every time we get a fresh package it must be run with root privileges as shown in image below. **sudo** is for root user.



```
(kali㉿kali)-[~/Desktop]
$ sudo apt update
[sudo] password for kali:
Get:1 http://kali.download/kali kali-rolling InRelease [34.0 kB]
Get:2 http://kali.download/kali kali-rolling/main amd64 Packages [21.4 MB]
Get:3 http://kali.download/kali kali-rolling/main amd64 Contents (deb) [53.3 MB]
Get:4 http://kali.download/kali kali-rolling/contrib amd64 Packages [114 kB]
Get:5 http://kali.download/kali kali-rolling/contrib amd64 Contents (deb) [265 kB]
Get:6 http://kali.download/kali kali-rolling/non-free amd64 Packages [183 kB]
Get:7 http://kali.download/kali kali-rolling/non-free amd64 Contents (deb) [915 kB]
Get:8 http://kali.download/kali kali-rolling/non-free-firmware amd64 Packages [15.8 kB]
Get:9 http://kali.download/kali kali-rolling/non-free-firmware amd64 Contents (deb) [40.5 kB]
Fetched 76.3 MB in 56s (1,373 kB/s)
1133 packages can be upgraded. Run 'apt list --upgradable' to see them.
```


7 TryHackme Linux Fundamentals Room

Linux is a core skill use in Daily life of cyber security work. To learn about linux TryHackme Linux Fundamentals part 1 is very good learning resource as it helped me to learn about commands such as listing files and folders, printing current working directory, viewing files, creating directory, creating files etc.



8 Conclusion

This week I started my Internship from Understanding the core terminologies of cyber security then moved to some networking concepts and then Setting up Isolated and Professional Lab environment for security assessment. This week serves as the base of building strong security skills in upcoming task.

